

Technology Stack

Automated Network Request Management Using ServiceNow

Project Name: Automated Network Request Management

Purpose: Define the technologies and ServiceNow components required to automate the complete lifecycle of network service requests, including request submission, workflow automation, approvals, task assignment, notifications, reporting, and security.

Layer / Component	Technology / Service	Purpose
Platform	ServiceNow	Central platform for managing network service requests, workflows, approvals, records, and administration.
User Interface	Service Catalog / Service Portal	Provides a self-service portal where employees submit network access requests by entering request details such as request type, device type, access level, business justification, and requested user.
Data Layer	Custom ServiceNow tables (Network Request)	Stores all network request information including request details, priority, approval status, assignment group, and request history.
Automation	Flow Designer	Retrieves catalog variables, validates request data, creates Network Request records, routes approvals, assigns tasks, and automates the complete workflow.
Business Logic	Business Rules / Script Includes	Implements server-side validation, business rules, reusable logic, and automation for network request processing.
Client-side Logic	Client Scripts / Catalog UI Policies	Validates user input, controls field behavior, enforces mandatory fields, and improves the user experience during request submission.
Notifications	ServiceNow Notifications	Sends automated notifications for request submission, approvals, rejections, assignments, and request completion.

Approval Management	Approval Engine	Routes requests to managers or network administrators for approval before processing.
Task Management	Flow Designer / Catalog Tasks	Automatically creates and assigns tasks to the appropriate IT support groups based on the request type.
Security	Roles, ACLs, Users, Groups	Protects application data, controls user access, and ensures only authorized users can view or modify network requests.
Reporting	ServiceNow Reports & Dashboards	Generates reports and dashboards to monitor request volume, approval status, pending requests, completed requests, and operational performance.
Integration	REST APIs / IntegrationHub (Future Enhancement)	Enables future integration with Active Directory, Azure AD, email services, identity management systems, and other enterprise applications.

Proposed Architecture

Employee → Service Catalog → Catalog Variables → Flow Designer → Network Request Record → Approval Engine → Task Assignment → Notifications → Reports & Dashboards → System Administrator.

Technology Selection Rationale

ServiceNow was selected because it provides a unified enterprise platform that combines **Service Catalog, custom tables, workflow automation, approvals, notifications, security, reporting, and dashboards** within a single environment. This eliminates manual request processing, reduces administrative effort, standardizes network request management, and improves operational efficiency. The platform also supports scalability through integrations, reusable business logic, and future enhancements such as REST APIs, AI-assisted automation, and identity management integration.

